S.No: 9	Exp. Name: *Fractional Knapsack Problem*	Date:2025-10-30

Aim:

The below program has a method void knapsack(). Which takes four parameters **number of objects**, the **weight of each object**, the **profit** corresponding to each one and the **capacity of the knapsack**. Write a program using a fractional knapsack algorithm to get the maximum profit.

Print the output as follows:

```
Sample Input and Output:
Enter the no. of objects: 6
Enter the weights and profits of each object:
1 2
4 5
8 9
4 6
5 2
3 5
Enter the capacity of knapsack:10
Maximum profit is:- 15.500000
```

Source Code:

knapsack.c

```
# include<stdio.h>
void knapsack(int n, float weight[], float profit[], float capacity) {
  // write your code here

   int i;
   float tot = 0.0;
   for(i=0; i<n; i++){
      if(capacity<=0){
         break;
      }
      if(weight[i]<=capacity){
         capacity -= weight[i];
         tot += profit[i];
      }
      else{
         tot += profit[i]*(capacity/weight[i]);
         break;
      }
   }
   printf("Maximum profit is:- %f\n",tot);





}
```


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```
int main() {
   float weight[20], profit[20], capacity;
   int num, i, j;
   float ratio[20], temp;
   printf("Enter the no. of objects: ");
   scanf("%d", &num);
   printf("Enter the weights and profits of each object:\n");
   for (i = 0; i < num; i++) {
      scanf("%f %f", &weight[i], &profit[i]);
   }
   printf("Enter the capacity of knapsack:");
   scanf("%f", &capacity);
   for (i = 0; i < num; i++) {
      ratio[i] = profit[i] / weight[i];
   }

   for (i = 0; i < num; i++) {
      for (j = i + 1; j < num; j++) {
         if (ratio[i] < ratio[j]) {
            temp = ratio[j];
            ratio[j] = ratio[i];
            ratio[i] = temp;
            temp = weight[j];
            weight[j] = weight[i];
            weight[i] = temp;
            temp = profit[j];
            profit[j] = profit[i];
            profit[i] = temp;
         }
      }
   }
   knapsack(num, weight, profit, capacity);
   return(0);
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter the no. of objects: 6
Enter the weights and profits of each object: 1 2
4 5
8 9
4 6
5 2
3 5
Enter the capacity of knapsack: 10
Maximum profit is:- 15.500000

Test Case - 2
User Output
Enter the no. of objects: 5
Enter the weights and profits of each object: 4 6

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1 3
7 5
5 3
3 4
Enter the capacity of knapsack: 10
Maximum profit is:- 14.428572